Card Production Solution Design

1. Solution Overview

The Card Production system processes eligible card requests in scheduled batches. The solution performs database processing, prepares card data, generates production files and sends completion information to downstream systems.

2. Main Components

The solution consists of the Batch Processor, Card Production Database, File Generation Service, Notification Service and downstream delivery integrations.

3. Batch Processor

The Batch Processor controls the execution of card-production batches. It reads eligible card records, performs processing steps and coordinates database updates and downstream activities.

4. Database Transaction Layer

Card-processing updates are written to the Card Production Database. Transaction boundaries are used to control commit and rollback behavior. Database deadlocks may occur when concurrent processing attempts to update conflicting records.

5. File Generation Service

After successful processing, the File Generation Service creates the card-production output file. A production batch should not create more than one valid output file for the same batch identifier.

6. Notification Service

After processing completes, the Notification Service sends completion information to downstream systems. Notifications must correspond to the final state of the batch.

7. Processing Sequence

The standard processing sequence is: 1. Select eligible card records. 2. Start batch processing. 3. Perform database updates. 4. Commit successful transaction processing. 5. Update batch status. 6. Generate the production file. 7. Deliver the file to the downstream integration. 8. Send completion notification.

8. Failure Handling

If processing fails before the database transaction commits, the transaction should be rolled back. If processing fails after a commit, restarting the same operation may repeat already completed work unless duplicate-prevention or idempotency controls are applied.

9. Retry Considerations

Database retry logic must distinguish between an operation that fully failed and an operation whose result may already have been committed. Repeated retries may increase database load and may also repeat downstream processing if retry boundaries include file generation or notification activities.

10. Dependencies

Card Production depends on the database, File Generation Service, Notification Service and downstream file-delivery integration. Failure in one component may leave the batch in a partially completed state and require reconciliation before processing continues.